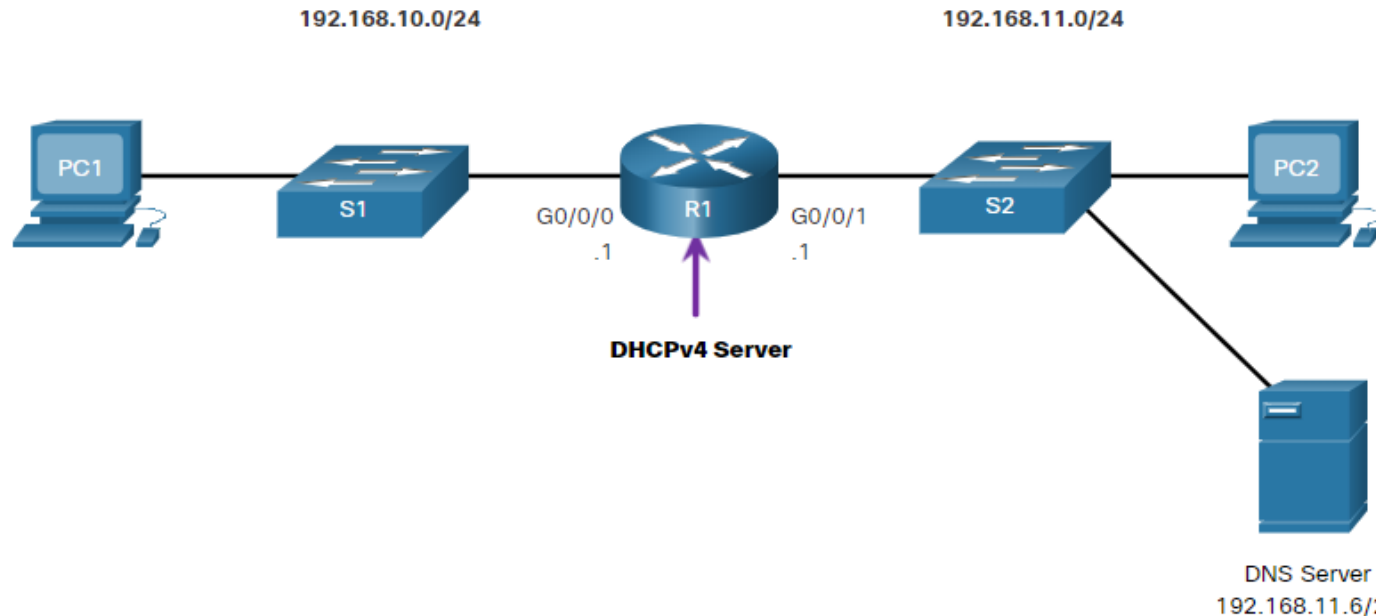


7.2 Configure a Cisco IOS DHCPv4 Server

Configure a Cisco IOS DHCPv4 Server

Cisco IOS DHCPv4 Server

- A Cisco router running Cisco IOS software can be configured to act as a DHCPv4 server.
- The Cisco IOS DHCPv4 server assigns and manages IPv4 addresses from specified address pools within the router to DHCPv4 clients.



Steps to Configure a Cisco IOS DHCPv4 Server

Use the following steps to configure a Cisco IOS DHCPv4 server:

- **Step 1.** Exclude IPv4 addresses.

The command is **ip dhcp excluded-address *low-address* [*high-address*]**

- **Step 2.** Define a DHCPv4 pool name.

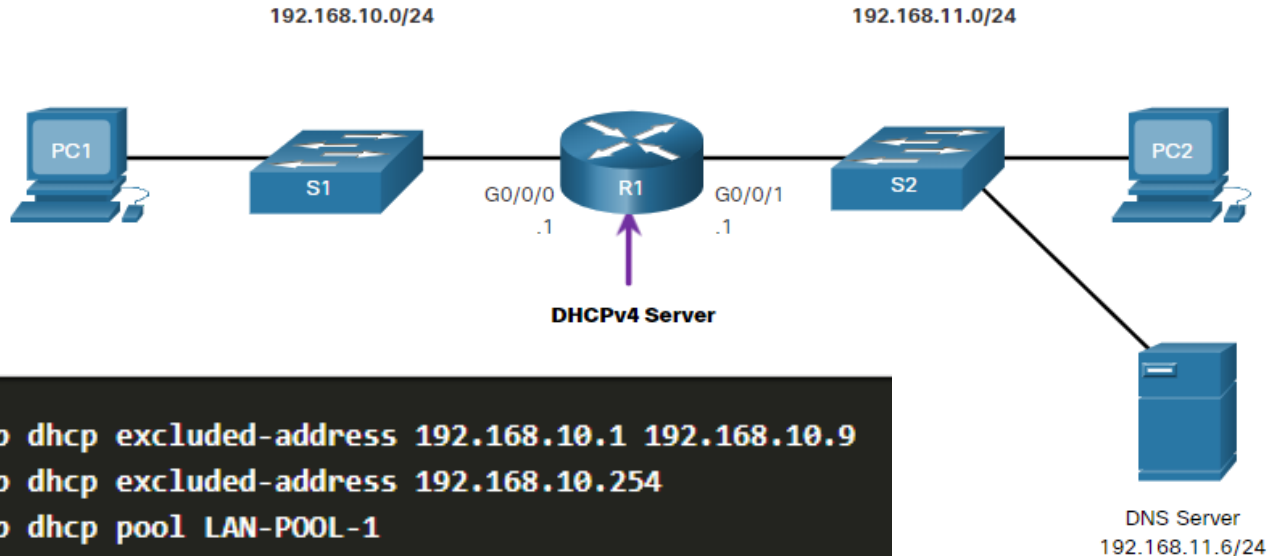
The **ip dhcp pool *pool-name*** command creates a pool with the specified name and puts the router in DHCPv4 configuration mode, which is identified by the prompt **Router(dhcp-config)#**.

Steps to Configure a Cisco IOS DHCPv4 Server (Cont.)

- **Step 3.** Configure the DHCPv4 pool. The address pool and default gateway router must be configured.
- Use the **network** statement to define the range of available addresses.
- Use the **default-router** command to define the default gateway router. These commands and other optional commands are shown in the table.

Task	IOS Command
Define the address pool.	network <i>network-number</i> [<i>mask</i> / <i>prefix-length</i>]
Define the default router or gateway.	default-router <i>address</i> [<i>address2</i> ... <i>address8</i>]
Define a DNS server.	dns-server <i>address</i> [<i>address2</i> ... <i>address8</i>]
Define the domain name.	domain-name <i>domain</i>
Define the duration of the DHCP lease.	lease { <i>days</i> [<i>hours</i> [<i>minutes</i>]] infinite }
Define the NetBIOS WINS server.	netbios-name-server <i>address</i> [<i>address2</i> ... <i>address8</i>]

Configure a Cisco IOS DHCPv4 Server Configuration Example



```
R1(config)# ip dhcp excluded-address 192.168.10.1 192.168.10.9
R1(config)# ip dhcp excluded-address 192.168.10.254
R1(config)# ip dhcp pool LAN-POOL-1
R1(dhcp-config)# network 192.168.10.0 255.255.255.0
R1(dhcp-config)# default-router 192.168.10.1
R1(dhcp-config)# dns-server 192.168.11.5
R1(dhcp-config)# domain-name example.com
R1(dhcp-config)# end
R1#
```

Configure a Cisco IOS DHCPv4 Server

DHCPv4 Verification

Use the commands in the table to verify that the Cisco IOS DHCPv4 server is operational.

Command	Description
show running-config section dhcp	Displays the DHCPv4 commands configured on the router.
show ip dhcp binding	Displays a list of all IPv4 address to MAC address bindings provided by the DHCPv4 service.
show ip dhcp server statistics	Displays count information regarding the number of DHCPv4 messages that have been sent and received

Configure a Cisco IOS DHCPv4 Server

Verify DHCPv4 is Operational

Verify the DHCPv4 Configuration: show running-config | section dhcp

command output displays the DHCPv4 commands configured on R1.

The **| section** parameter displays only the commands associated with DHCPv4 config

```
R1# show running-config | section dhcp
ip dhcp excluded-address 192.168.10.1 192.168.10.9
ip dhcp excluded-address 192.168.10.254
ip dhcp pool LAN-POOL-1
  network 192.168.10.0 255.255.255.0
  default-router 192.168.10.1
  dns-server 192.168.11.5
  domain-name example.com
```

Configure a Cisco IOS DHCPv4 Server

Verify DHCPv4 is Operational (Cont.)

Verify DHCPv4 Bindings:, the operation of DHCPv4 can be verified using the **show ip dhcp binding** command.

This command displays a list of all IPv4 address to MAC address bindings that have been provided by the DHCPv4 service.

```
R1# show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type      State      Interface
                Hardware address/
                User name
192.168.10.10    0100.5056.b3ed.d8  Sep 15 2019 8:42 AM  Automatic Active
GigabitEthernet0/0/0
```

Configure a Cisco IOS DHCPv4 Server

Verify DHCPv4 is Operational (Cont.)

Verify DHCPv4 Statistics: The output of the **show ip dhcp server statistics** is used to verify that messages are being received or sent by the router.

This command displays count information regarding the number of DHCPv4 messages that have been sent and received.

```
R1# show ip dhcp server statistics
```

Memory usage	19465
Address pools	1
Database agents	0
Automatic bindings	2
Manual bindings	0
Expired bindings	0
Malformed messages	0
Secure arp entries	0
Renew messages	0
Workspace timeouts	0
Static routes	0
Relay bindings	0
Relay bindings active	0
Relay bindings terminated	0
Relay bindings selecting	0
Message	Received
BOOTREQUEST	0
DHCPDISCOVER	4
DHCPREQUEST	2
DHCPDECLINE	0
DHCPRELEASE	0
DHCPINFORM	0

Configure a Cisco IOS DHCPv4 Server

Verify DHCPv4 is Operational (Cont.)

Verify DHCPv4 Client Received IPv4 Addressing:

The **ipconfig /all** command, when issued on PC1, displays the TCP/IP parameters, as shown in the example

```
C:\Users\Student> ipconfig /all
Windows IP Configuration

Host Name . . . . . : ciscolab
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : example.com
    Description . . . . . : Realtek PCIe GBE Family Controller
    Physical Address. . . . . : 00-05-9A-3C-7A-00
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    IPv4 Address. . . . . : 192.168.10.10
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained . . . . . : Saturday, September 14, 2019 8:42:22AM
    Lease Expires . . . . . : Sunday, September 15, 2019 8:42:22AM
    Default Gateway . . . . . : 192.168.10.1
    DHCP Server . . . . . : 192.168.10.1
    DNS Servers . . . . . : 192.168.11.5
```

Disable the Cisco IOS DHCPv4 Server

The DHCPv4 service is enabled by default.

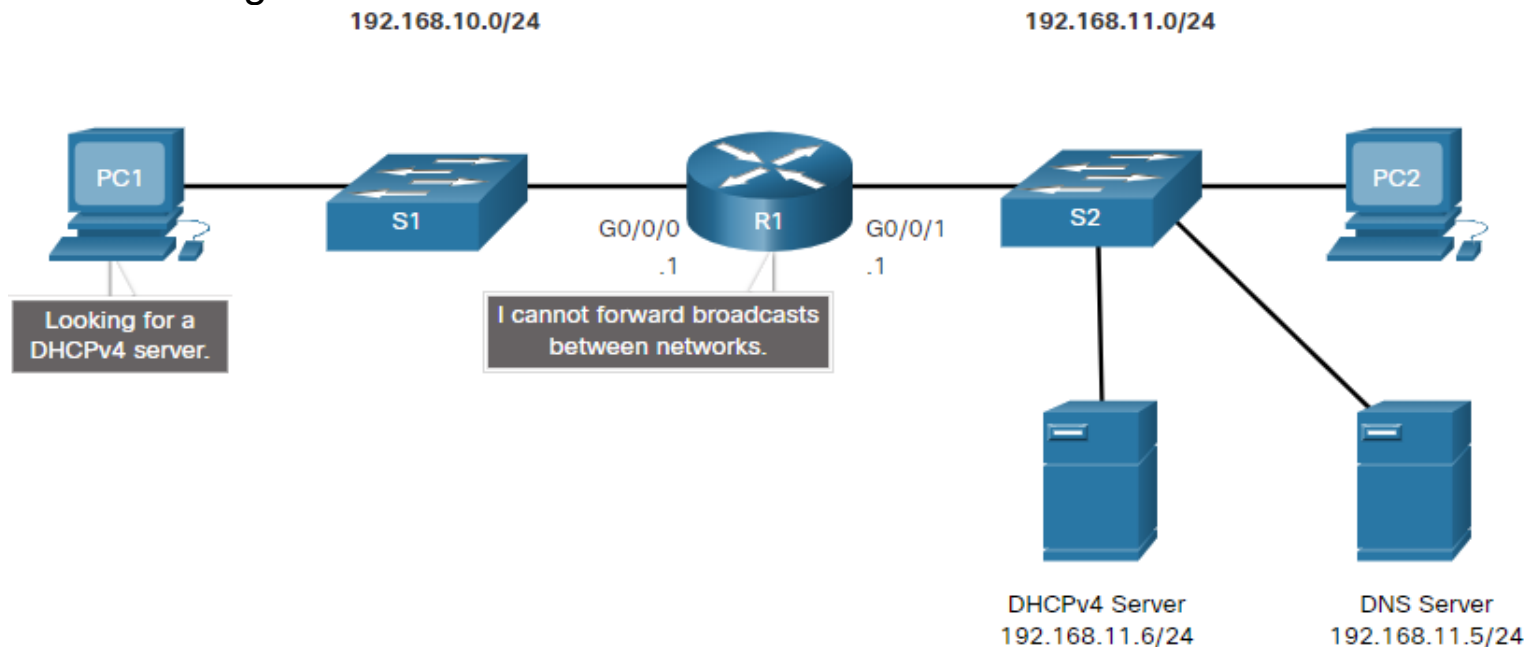
To disable the service, use the **no service dhcp** global configuration mode command

```
R1(config)# no service dhcp
R1(config)# service dhcp
R1(config)#
```

Configure a Cisco IOS DHCPv4 Server

DHCPv4 Relay

- In a complex hierarchical network, enterprise servers are usually located centrally. In the figure, PC1 is attempting to acquire an IPv4 address from a DHCPv4 server using a broadcast message



Configure a Cisco IOS DHCPv4 Server

DHCPv4 Relay (Cont.)

- Configure R1 with the **ip helper-address** *address* interface configuration command.
- When R1 has been configured as a DHCPv4 relay agent, it accepts broadcast requests for the DHCPv4 service and then forwards those requests as a unicast to the IPv4 address 192.168.11.6..

```
R1(config)# interface g0/0/0
R1(config-if)# ip helper-address 192.168.11.6
R1(config-if)# end
R1#
```

```
R1# show ip interface g0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
Internet address is 192.168.10.1/24
Broadcast address is 255.255.255.255
Address determined by setup command
MTU is 1500 bytes
Helper address is 192.168.11.6
(output omitted)
```

Configure a Cisco IOS DHCPv4 Server

Other Service Broadcasts Relayed

DHCPv4 is not the only service that the router can be configured to relay. By default, the **ip helper-address** command forwards the following eight UDP services:

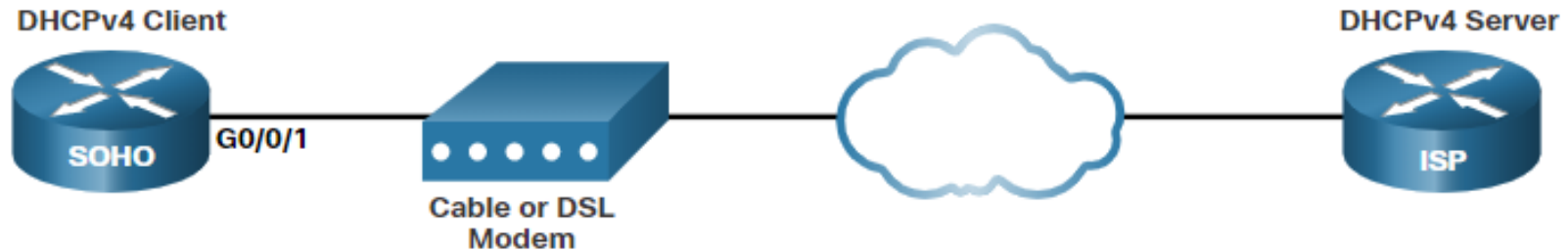
- Port 37: Time
- Port 49: TACACS
- Port 53: DNS
- Port 67: DHCP/BOOTP server
- Port 68: DHCP/BOOTP client
- Port 69: TFTP
- Port 137: NetBIOS name service
- Port 138: NetBIOS datagram service

7.3 Configure a DHCPv4 Client

Configure a DHCPv4 Client Cisco Router as a DHCPv4 Client

There are scenarios where you might have access to a DHCP server through your ISP.

In these instances, you can configure a Cisco IOS router as a DHCPv4 client.



Configure a DHCPv4 Client

Configuration Example

- To configure an Ethernet interface as a DHCP client, use the **ip address dhcp** interface configuration mode command,

```
SOHO(config)# interface G0/0/1
SOHO(config-if)# ip address dhcp
SOHO(config-if)# no shutdown
Sep 12 10:01:25.773: %DHCP-6-ADDRESS_ASSIGN: Interface GigabitEthernet0/0/1 assigned DHCP address
209.165.201.12, mask 255.255.255.224, hostname SOHO
```

```
SOHO# show ip interface g0/0/1
GigabitEthernet0/0/1 is up, line protocol is up
  Internet address is 209.165.201.12/27
  Broadcast address is 255.255.255.255
  Address determined by DHCP
(output omitted)
```